
OncoScreen: A Random-Forest-Benchmarked, Logistic-Regression-Deployed Model for Breast Cancer Disease Prediction

CodeAlpha Machine Learning Internship — Task 4 (2 of 2 tasks completed)

SUMMARY

- Trained on the real Wisconsin Diagnostic Breast Cancer dataset (569 samples, 30 features).
- Primary benchmark: Random Forest, 30 features — 94.7% accuracy, 0.994 ROC-AUC.
- Deployed model: Logistic Regression distilled to the top-8 RF-ranked features for browser transparency.
- Deployed model nearly matches the full benchmark: 94.7% accuracy, 0.993 ROC-AUC on just 8 features.
- Full metric suite reported: Precision, Recall, F1-Score, and ROC-AUC as the brief specifies.
- Interactive HTML tool with preset benign/malignant samples and a live risk gauge.

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1. Abstract

This paper documents Task 4 of the CodeAlpha Machine Learning internship: disease prediction from structured medical data. The brief specifically names the Breast Cancer dataset and suggests SVM, Logistic Regression, Random Forest, or XGBoost. This implementation trains a Random Forest on the real, UCI-sourced Wisconsin Diagnostic Breast Cancer dataset as the primary benchmark, then distills a transparent Logistic Regression on the eight most important features for deployment in a live browser tool, reporting the full Precision/Recall/F1/ROC-AUC suite the brief requests.

2. Dataset

569 samples (357 benign, 212 malignant), 30 real-valued features describing cell-nuclei measurements from digitized fine-needle-aspirate images, distributed unmodified via scikit-learn's bundled loader (mirroring the canonical UCI repository version).

3. Model Results

Model	Accuracy	Precision	Recall	F1	ROC-AUC
Random Forest (30 features)	94.7%	95.8%	95.8%	95.8%	0.994
Deployed Logistic Regression (8 features)	94.7%	97.1%	94.4%	95.8%	0.993

The eight-feature deployed model matches the full 30-feature benchmark almost exactly on every metric. All eight selected features come from either the 'worst' or 'mean' measurement group for size (radius/perimeter/area) or shape irregularity (concave points), consistent with established clinical intuition that larger, more irregularly-shaped nuclei indicate malignancy.

4. Deployed Model

The browser tool reimplements the exact standardize-then-sigmoid formula from the exported scaler and Logistic Regression coefficients in JavaScript, with preset benign-like and malignant-like samples and a live risk gauge, plus a persistent non-diagnostic disclaimer since this is an educational deliverable, not a clinical device.

5. Clinical Framing

Recall is highlighted alongside Precision because in a screening context a false negative (a missed cancer) is typically far more costly than a false positive, and the brief's full metric suite is reported rather than accuracy alone, which can mask this asymmetry.

6. Limitations

The dataset is single-institution and dates to early-1990s imaging methodology; a production clinical model would need validation on contemporary, multi-institution data, and this tool is explicitly not a diagnostic device.

7. Conclusion

OncoScreen satisfies the brief using a real UCI benchmark dataset, the full requested metric suite, and a distilled 8-feature model that matches its 30-feature benchmark almost exactly while remaining fully auditable in the browser.

References

- [1] Wolberg, W.H., Street, W.N., Mangasarian, O.L. Breast Cancer Wisconsin (Diagnostic) Data Set. UCI ML Repository.
- [2] CodeAlpha. (2026). Machine Learning Tasks & Instructions. www.codealpha.tech.

“The best clinical model is the one whose every weight you can still explain.”

— **Mayank Swaraj**